

Deterministic Optimization

Geometric Aspects of Linear Optimization

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Basic Geometric Objects in LP

Geometric Aspects of Linear Optimization

Learning Objectives for this module

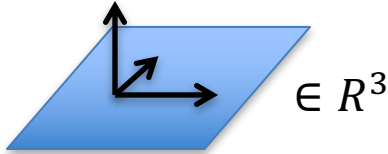
- Discover the interrelation between geometric and algebraic aspects of linear optimization
- Use geometric language to study LP

Geometric Objects and Algebraic Forms

- Points and vectors
 - $x \in R^n$
- Rays, lines, and their parametric forms



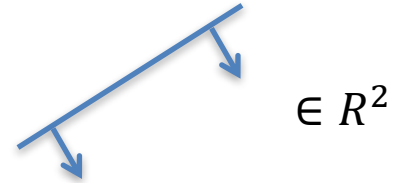
- Plane and solutions of linear equations



- Hyperplane in higher dimension



- Halfspace and one linear inequality



- Polyhedron and “several” linear inequalities



Points and Vectors

- A point is a geometric object in space
- Algebraically, a point in n -dimensional space is given by its coordinates: $\boldsymbol{x} = (x_1, \dots, x_n)^\top \in R^n$
- In this way, a point is also called a vector
- We always write a vector as a column vector

Rays, Lines, and Parametric Form

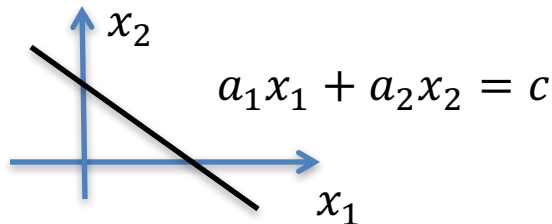
- A ray consists of a starting point a and all the points in a direction d



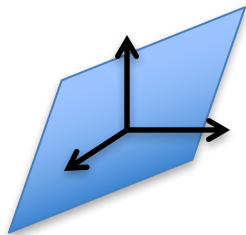
- How do we describe a ray in algebraic form?
 - $\{x \in R^n \mid x = a + \theta d, \forall \theta \geq 0\}$
- A line consists of two rays starting at a point pointing two opposite directions
 - $\{x \in R^n \mid x = a + \theta d, \forall \theta \in R\}$
 - There is another way to represent a line.

Plane and A Linear Equation

- A plane in R^2 is just a line.



- A plane in R^3

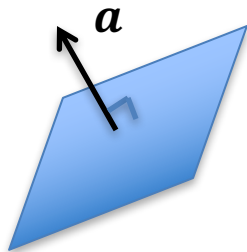


$$a_1x_1 + a_2x_2 + a_3x_3 = c$$

If $c = 0$, plane passes through original.

Hyperplane and A Linear Equation

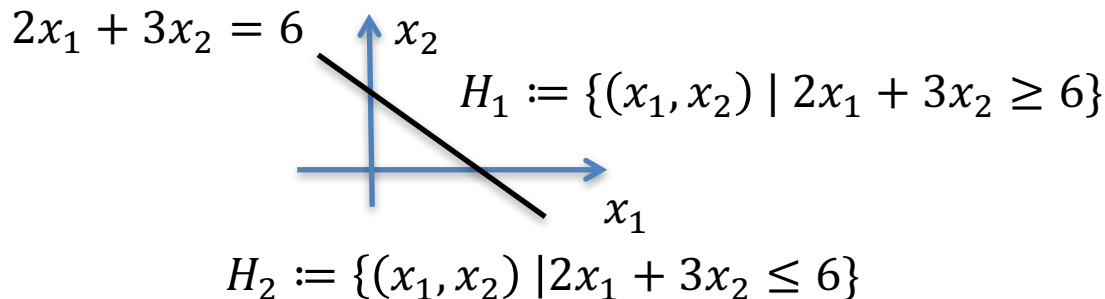
- The concept of a plane can be extended to any dimension R^n
- Algebraically, a plane in R^n is the set of solutions of one linear equation in n variables
 - $a_1x_1 + \cdots + a_nx_n = c$
- We can write it in compact form: $\mathbf{a}^\top \mathbf{x} = c$



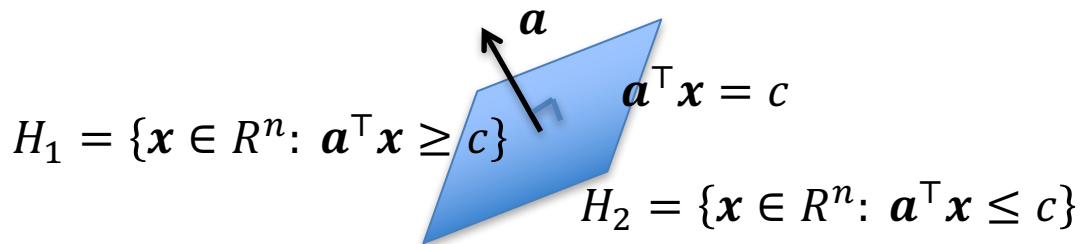
$\mathbf{a}$: normal vector

Halfspace and a linear inequality

- In R^2 , a halfspace is half of the whole space

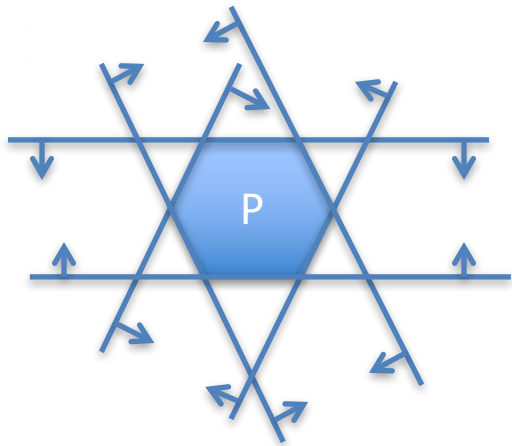


- Same definition can be extended to a halfspace in R^n



Polyhedron and several halfspaces

- Definition: A **polyhedron** is the **intersection** of a **finite number of halfspaces**



$$P = \{x \in R^n \mid a_1^T x \leq b_1, \\ \dots \\ a_m^T x \leq b_m\} \\ = \{x \in R^n \mid Ax \leq b\}$$

$$A = \begin{bmatrix} a_1^T \\ \vdots \\ a_m^T \end{bmatrix} \quad b = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

P is exactly the feasible region of an LP.

Summary

- We described some basic geometric objects by algebraic forms
- We built up the geometric elements for a linear program
 - Point
 - Line
 - Plane
 - Hyperplane
 - Halfspace
 - Polyhedron